

Pratik Bhanuse

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PROFESSIONAL SUMMARY AND SKILLS

Quantitative finance & ML specialist with a background in theoretical and computational physics, experienced in systematic trading, financial modeling, and feature engineering. Expert in GNNs, LSTMs, and Transformers for asset allocation, regime detection, and risk control. Proven ability to fuse macroeconomic, fundamental, and sentiment data into interpretable, data-driven strategies. Highly proficient in Python, C++, and SQL, adept at turning complex research into actionable insights.

EDUCATION

Doctor of Philosophy (Ph.D.) Physics, The University of Western Ontario, Canada. Sept 2024 – Present
Theoretical and Computational Physics
Supervisor: Prof. Colin Denniston

Master of Science (M.Sc.) Materials Physics, Brock University, Canada. Aug 2019 – Dec 2020
GPA: 3.8/4.0
Project: Synthesis and characterization of ReMo and LSMO superconductors.

Master of Science (M.Sc.) Theoretical Physics, The University of Nottingham, UK Sept 2015 – Dec 2016
Thesis: "Coherent States in String theory"
Recipient of "Developing Solutions Scholarship"

Bachelor of Science (B.Sc.) Physics, University of Mumbai, India. July 2012 – May 2015
Recipient of "Certificate of Excellence" for conference on sustainable development from United Nations.

Programming: Python, NumPy, pandas, scikit-learn, PyTorch, TensorFlow, C++, Cython, SQL, q/kdb+, Bash/Shell scripting, API design, object-oriented programming

Data Science & Machine Learning: Supervised/unsupervised learning, time-series modelling (LSTM, CNN, Transformers), graph-based methods (GNN), dimensionality reduction (PCA, TDA), data structures, pattern recognition, feature engineering

Quantitative Finance & Risk Modelling: Algorithmic trading, statistical arbitrage, pairs trading, portfolio optimization, PD/LGD/EAD modelling, stress testing, VaR, regime detection, slippage control, Equities, ETFs, bonds, commodities, FX, derivatives, OTC markets

WORK EXPERIENCE

Quantitative Strategy Analyst, BCA Research, Montreal, CA Mar 2022 – Mar 2025

- Built macro-driven asset-allocation models that diversified portfolios across equities, bonds, and cash, and collaborating with other strategists.
- Analyzed and sanitized multi-terabyte financial datasets to support systematic trading model development.
- Expertise in monitoring, evaluating, and enhancing the behavior of existing models while incorporating cutting-edge techniques in quantitative finance.
- Contributing to the production of original research publications linking macro themes to equity factor models, with a focus on high-quality visuals for publications and client presentations.
- Providing advanced support to clients accessing model signals through digital platforms, ensuring seamless user experiences, and delivering exceptional client service.

- Risk Analyst – AML/ATF and compliance, *AML RightSource, Mississauga, CA*** Feb 2021 – Mar 2022
- Built a hybrid financial crime detection system combining interpretable Random Forests with deep learning models like LSTM and GNNs.
 - Analysed transaction data across banks and MSBs to detect fraud, money laundering, and suspicious activity patterns.
 - Flagged high-risk accounts using ensemble classifiers and relational modelling of transactional graphs for real-time monitoring.
 - Filed SARs and conducted OFAC, US, Canadian, and international sanctions screening to support regulatory compliance decisions.

- Quantitative analyst internship, *QTS Capital Management, LLC., Niagara, CA*** July 2020 – Feb 2021
Under the supervision of Dr. Ernest Chan (aka Ernie - <https://epchan.com/>)
- Designed statistical arbitrage models using cointegration-based pairs trading across ETFs, indices, and equities.
 - Enhanced trading strategies with ML-driven signals using features like momentum, volatility, and mean-reversion.
 - Applied FinBERT-based NLP to classify company sentiment from annual and quarterly reports.
 - Integrated sentiment analysis into asset screening to improve trade selection within statistical arbitrage frameworks.

- Research Analyst, *Indian Institute of Technology - Bombay (IIT-B), Mumbai India*** Feb 2017 – Aug 2017
- Performed Python- and Mathematica-based data analysis, regression, and ML-driven simulations to optimize nanomaterial synthesis for cancer drug delivery.
 - Authored technical reports and presented progress updates to faculty and research team members.

LEADERSHIP INITIATIVES

- President, *Phynance Association – Brock University.*** Jan 2020 – Dec 2020
- Founded and led the Phynance Association at Brock University, uniting physics, machine learning, and quantitative finance.
 - Curated a lecture series by faculty and industry experts on practical quantitative finance applications.
 - Organized hands-on workshops and algorithmic trading simulations to deepen members' real-world skills.
 - Directed PR, marketing, sponsorship, and recruitment to drive engagement and community growth.

- Graduate Student Association Representative, *Space Evaluation Committee*** Oct 2019 – Dec 2020
- Reviewed and evaluated space allocation requests from faculty and students to ensure alignment with institutional objectives and approved space-management policies
 - Conducted detailed campus space audits to assess current utilization and identify areas suitable for renovation or expansion - supporting planning decisions for new divisions and department upgrades.

CERTIFICATION

- Applied Data Science I: Scientific Computing & Python (with honors) – WorldQuant University
- Bloomberg Market Concepts (BMC) – Bloomberg LP